Hannah Brake

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portfolio | linkedin | github

Education

B. Eng. Computer Engineering — **Sept. 2017 - Apr. 2022**

Memorial University of Newfoundland and Labrador — *GPA: 3.77/4.0 (83% Avg)*

Work Experience

Software Test Engineer — **Jun. 2022 - Present**

Nokia — *Ottawa, Ontario*

- Developing **test automation** using **Python/Pytest**, VS Code, Git and hardware test benches in a **Linux** based regression environment to conduct **system-level testing** of Nokia's 7750, 7250 and 7730 router products to ensure high product quality.
- Executing **end-to-end test cycles** for Precision Time Protocol based hardware and software features. Participating in **review of requirements documents** and comprehensive **test plan** development.
- Providing detailed **reporting of software defects** in addition to ensuring **defect resolution** through verification testing and collaborating with software, hardware and product management teams.
- Performing **triaging of regression test failures** and correcting failures through **debugging**, communicating with development teams and performing **test enhancements**.
- Assisting with **hiring** and talent attraction activities including **resume screening**, **interviewing** and final candidate selection for new software test co-ops and full-time employees. **Training and mentoring** co-ops and newly hired team members on software test tools, enviroment and processes.

Embedded Systems Developer Co-op — **Sept. 2019 - Dec. 2019**

Memorial University of Newfoundland - Makerspace — *St. John's, Newfoundland*

- Developed a security **checkpoint device prototype** using a **Raspberry Pi 3**, 16x2 GPIO LCD screen, **Python**, **MySQL** database and additional **embedded** hardware components.
- Supervised a team of employees and conducted educational technology workshops on both embedded **Arduino** development and **Unity/C#** video game development.

Projects & Initiatives

| Killick-1 CubeSat

Mission Control System Developer

- Worked in a team of five for a Canadian Space Agency initiative to develop an **I2C based Mission Control Software** for the Memorial University **cube satellite**.
- Created a custom API written in **C** to manage and monitor voltage rails/power distribution to other subsystem hardware modules. Developed **platform test** scripts and performed Hardware-in-the-Loop testing via a custom "flat-sat" board to ensure correct API functionality.
- Utilized **Code Composer Studio**, and embedded modules (ABACUS Satellite On-Board Computer and ISISPACE Compact Electrical Power System) for development.
- Wrote **technical documentation** depicting all hardware/software testing conducted, C development and tools used for the current phase of the project, in addition to a comprehensive overview of future project plans for the CubeSat. Formally presented this to representatives from the Canadian Space Agency in a PowerPoint format.

| iOS Weather App

- Displays daily **weather information** for the current location, along with a seven-day forecast. **Swift**, **Xcode IDE** and the **OpenWeatherMap API** were used for development.

| XOR Encryption Cipher

- Developed an **XOR Cipher** using **Java** (and a Swing based GUI), which prompts a user to enter a password and key value. An encrypted message is produced.

| Ottawa Career Fair Enthusiast

- Selected to represent Nokia at the **University of Ottawa** and **Kanata North career fairs** in 2023. Networked with 200+ students, fostered engagement on 5G innovations and answered questions on software testing.

Skills

Programming Languages	Python | TCL | Bash | MATLAB | C++/C | C# | VHDL | Assembly | Java | Swift
Tools	Oscilloscope | Raspberry Pi | Arduino IDE | VS Code | MobaXTerm | Confluence
Technologies	Linux | Git | FreeRTOS | Altera FPGA | ModelSim | Pytest | MySQL | Wireshark
Protocols	I2C | GPIO | UART | SPI | TCP/UDP | IP | PTP
Strengths	problem solving | planning | attention to detail | motivator & leader